

Elastic Topology Management for Hybrid Retrieval-Augmented Generation: A Measurement Study across Edge, LAN, and WAN

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1 Background and Current State

We have built and deployed a working *hybrid* Retrieval-Augmented Generation (RAG) system: sparse keyword retrieval (BM25 via Tantivy), dense semantic retrieval (BGE-M3 embeddings searched in Qdrant), reciprocal-rank fusion, and LLM generation (Llama-3-8B-Instruct, AWQ 4-bit, served by vLLM [14, 13]). The corpus is the full MS MARCO passage collection [12]: **8,841,823 passages** embedded at 1024 dimensions, HNSW-indexed in Qdrant, ingested at roughly 538 passages per second on an RTX 3090. The system is deployed on two LAN machines connected over gRPC (WireGuard): a gateway node (embedding, dense+sparse retrieval, fusion) on an Intel i7-10750H laptop with GTX 1660 Ti, and a generation node (document lookup, vLLM) on an RTX 3090 host. An earlier three-node version placed a third, low-power laptop (Intel CPU, 16 GB RAM) at the network edge as entry point and sparse-retrieval engine, across a city-wide link treated as the WAN leg.

2 A Failed Hypothesis That Points at a Real Gap

The original design hypothesized a 30–40% TTFT (time-to-first-token) reduction by running edge-side sparse retrieval in parallel with transit of the query toward the other nodes: $T_{\text{retrieval}} = \max(T_{\text{sparse}}, T_{\text{WAN}} + T_{\text{dense}})$. Our instrumented three-node runs showed nothing like it. The surviving 100-query log records sparse retrieval at 0.6–10 ms against TTFT of 51–227 ms: the stage we chose to distribute was one to two orders of magnitude cheaper than the pipeline it was meant to accelerate. Since parallel overlap can save at most $\min(T_{\text{sparse}}, T_{\text{WAN}} + T_{\text{dense}})$, no link latency could have delivered the target — a short WAN leg offered little to hide, and a long one would only have slowed the dense branch. One caveat: we can no longer reconstruct where the machines physically were during those logged runs (all on one LAN, or the edge laptop across town), so we treat the 5–15 ms round-trip in our mid-evaluation report as a descriptive figure, not one tied to this log — and the new campaign will record placement and network metadata with every measurement. The three-node setup itself ended for operational reasons, not hardware: the edge laptop belonged to a groupmate who was often unavailable, so we consolidated its duties into the gateway and the two-node system worked fine.

That failure motivates this proposal: placement by intuition silently costs performance, the stages worth distributing are the *expensive* ones (dense retrieval, document handling, generation — not sparse lookup), and topology change is a normal event in real deployments — yet the literature, as far as we can find, cannot answer: *where should each stage of a hybrid RAG pipeline run, under which network conditions, and what exactly do we lose when the topology degrades?*

3 The Gap in the Literature

DNN partitioning between device and cloud is well studied at the *layer* level (Neurosurgeon [1], PipeEdge [2]); LLM serving splits model *phases* across machines (Splitwise [3]). For RAG specifically, existing work optimizes KV caching of retrieved knowledge (RAGCache [4], TurboRAG [5]), serving-system optimization (RAGO [6], METIS [7]), index management on a single host (EdgeRAG [8], VectorLiteRAG [9]), or data distribution across edge/cloud tiers (EACO-RAG [10], DGRAG [11]). We could not find a measurement study and placement policy for *which pipeline stage runs on which machine* in a multi-node hybrid RAG deployment: the five stages (sparse, dense, fusion, hydration, generation) can each sit on a different node, and the right answer changes with RTT, bandwidth, and node capability. Node loss (what happens to latency and answer quality when a node drops out

and roles consolidate) appears entirely unmeasured for RAG.

4 Research Questions

- **RQ1 (Placement).** Which node placement for each pipeline stage minimizes TTFT and end-to-end latency under a given network regime (RTT, bandwidth), and where does the best placement change?
- **RQ2 (Degradation).** When a node drops out — hardware failure or simple unavailability (our real three-to-two consolidation) — what is the cost in latency and retrieval quality, and how fast can the system adapt?
- **RQ3 (Policy).** Can a simple analytical cost model — per-stage compute times, payload sizes, network terms — predict the best placement well enough to drive decisions without the full experiment matrix?

5 Method

Testbed. The existing heterogeneous nodes (RTX 3090 host, GTX 1660 Ti laptop, revived low-power laptop) over WireGuard; the full 8.84M-passage index is already in place. **Instrumentation.** We will extend the existing per-stage timing logs into a structured benchmark harness (per-stage milliseconds, TTFT, decode throughput, true stream end-to-end) and add retrieval-quality evaluation against MS MARCO qrels (recall@ k), which the system does not yet measure; the sparse index will be scaled from a small test index to the full corpus. **Network control.** Per-link netem sweeps of delay and bandwidth (e.g. 0/10/40/80 ms RTT) make the WAN regime reproducible; key configurations are re-validated over the genuine WAN link to the edge laptop. **Placement matrix.** Stage-to-node assignment becomes a configuration switch (fusion on gateway vs. generator; hydration local vs. remote; sparse retrieval at the edge vs. LAN), reconstructing the archived three-node code paths. **Campaign.** A factorial sweep over placement $\times$ network regime $\times$ retrieval depth with repeats and percentile statistics; a degradation experiment kills nodes mid-load to measure failover cost; the RQ3 cost model is fit on half the configurations and validated on the held-out half.

6 Expected Contributions

(1) To our knowledge, the first factorial measurement of stage-to-node placement for hybrid RAG on commodity heterogeneous hardware under controlled WAN regimes; (2) a measurement of what topology degradation actually costs, grounded in our own three-to-two-node consolidation; (3) a validated cost-model-based placement policy; (4) an open, reproducible testbed with all artifacts.

7 Timeline and Target Venues

Weeks 1–2: harness, full-corpus sparse index, quality evaluation. Week 3: placement matrix and network emulation. Weeks 4–5: campaigns, cost model, analysis. Weeks 6–8: writing, internal review, submission. We target Q1 journals with rolling submission: *Future Generation Computer Systems*, *Computer Networks*, or *IEEE Transactions on Cloud Computing* (all currently Q1 in SCImago/JCR rankings). A conference extension (e.g. IEEE ICDCS 2027, call expected around December 2026) remains possible for the policy contribution.

References

- [1] Y. Kang et al., “Neurosurgeon: Collaborative Intelligence Between the Cloud and Mobile Edge,” *Proc. ASPLOS*, 2017.
- [2] Y. Hu et al., “PipeEdge: Pipeline Parallelism for Large-Scale Model Inference on Heterogeneous Edge Devices,” *Proc. DSD*, 2022.
- [3] P. Patel et al., “Splitwise: Efficient Generative LLM Inference Using Phase Splitting,” *Proc. ISCA*, 2024.
- [4] C. Jin et al., “RAGCache: Efficient Knowledge Caching for Retrieval-Augmented Generation,” arXiv:2404.12457, 2024.
- [5] S. Lu et al., “TurboRAG: Accelerating Retrieval-Augmented Generation with Precomputed KV Caches for Chunked Text,” *Proc. EMNLP*, 2025.
- [6] W. Jiang et al., “RAGO: Systematic Performance Optimization for Retrieval-Augmented Generation Serving,” *Proc. ASPLOS*, 2025.
- [7] S. Ray et al., “METIS: Fast Quality-Aware RAG Systems with Configuration Adaptation,” arXiv:2412.10543, 2025.
- [8] K. Seemakhupt, S. Liu, and S. Khan, “EdgeRAG: Online-Indexed RAG for Edge Devices,” arXiv:2412.21023, 2024.
- [9] J. Kim and D. Mahajan, “VectorLiteRAG: Latency-Aware and Fine-Grained Resource Partitioning for Efficient RAG,” arXiv:2504.08930, 2025.
- [10] J. Li et al., “EACO-RAG: Towards Distributed Tiered LLM Deployment Using Edge-Assisted and Collaborative RAG with Adaptive Knowledge Update,” arXiv:2410.20299, 2024.
- [11] W. Zhou, Y. Yan, and Q. Yang, “DGRAG: Distributed Graph-based Retrieval-Augmented Generation in Edge-Cloud Systems,” arXiv:2505.19847, 2025.
- [12] P. Bajaj et al., “MS MARCO: A Human Generated MACHine Reading COMprehension Dataset,” arXiv:1611.09268, 2016.
- [13] J. Chen et al., “M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation,” arXiv:2402.03216, 2024.
- [14] W. Kwon et al., “Efficient Memory Management for Large Language Model Serving with PagedAttention,” *Proc. SOSP*, 2023.